SCHOOL OF COMPUTER SCIENCE ENGINEERING

AND INFORMATION SYSTEMS

Case Study -M.TECH (SE) – Winter Semester - 2025-26

TITLE OF THE PROJECT

INVOICE SYSTEM WORKFLOW SYSTEM

AGILE DEVOPS AND DEVELOPMENT PROCESS

SURESH.I [23MIS0435]

65. Invoice Generation System Problem Statement:

Ensure invoices are generated accurately with correct billing details.

Tasks:

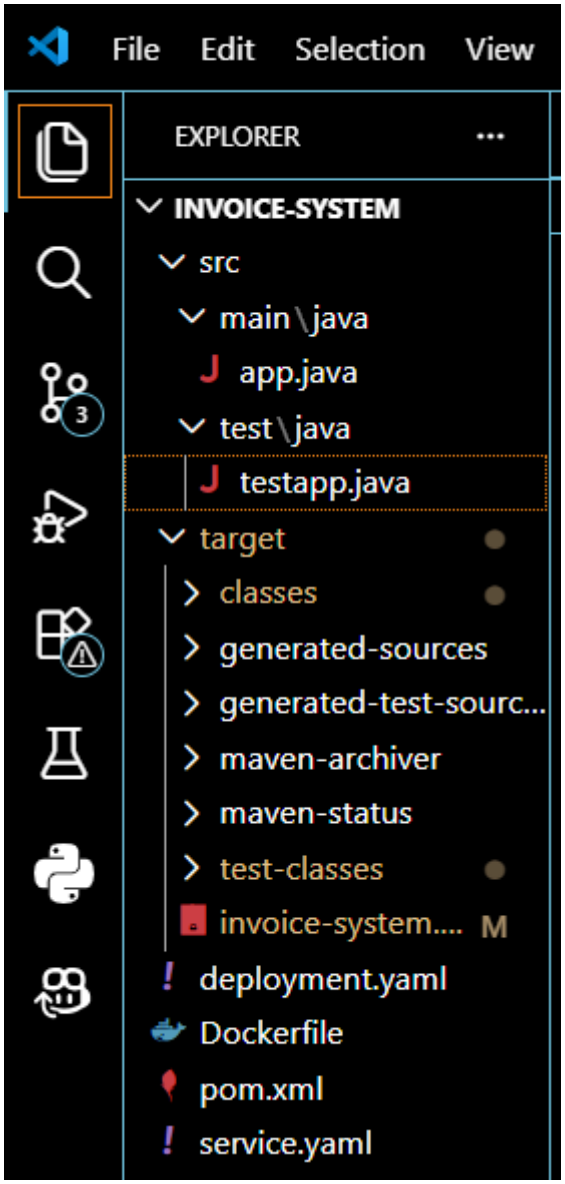
• Develop invoice generation logic

• Write JUnit tests for: o Billing accuracy o Tax and discount calculations

• CI/CD pipeline for automated deployment

• Deploy on Kubernetes Real-world relevance: Financial systems

FILE STRUCTURE:



1)APP.JAVA CODE:

```
public class app {


    // calculate amount

    public double calculateAmount(double price, int quantity) {

        return price * quantity;

    }


    // apply discount

    public double applyDiscount(double amount, double percent) {

        return amount - (amount * percent / 100);

    }


    // apply tax (GST)

    public double applyTax(double amount, double percent) {

        return amount + (amount * percent / 100);
```

```
    }

}
```

2)TESTAPP.JAVA

```
import java.util.Scanner;


public class testapp {


    public static void main(String[] args) {


        Scanner sc = new Scanner(System.in);

        app hotel = new app();


        System.out.println("============ HOTEL INVOICE SYSTEM ============");


        // Customer Details

        System.out.print("Enter Customer Name: ");

        String customerName = sc.nextLine();
```

```
System.out.print("Enter Table Number: ");

int tableNo = sc.nextInt();



System.out.print("Enter number of items ordered: ");

int n = sc.nextInt();



double grandTotal = 0;



System.out.println("\n---------- ORDER DETAILS ----------");



for (int i = 1; i <= n; i++) {



    sc.nextLine(); // clear buffer



    System.out.println("\nItem " + i);



    System.out.print("Enter item name: ");

    String itemName = sc.nextLine();
```

```
    System.out.print("Enter price: ");
    double price = sc.nextDouble();

    System.out.print("Enter quantity: ");
    int quantity = sc.nextInt();

    double amount = hotel.calculateAmount(price, quantity);
    grandTotal += amount;

    System.out.println("Item: " + itemName +
                       " | Qty: " + quantity +
                       " | Amount: " + amount);
}

// Discount & GST
System.out.print("\nEnter Discount (%): ");
double discount = sc.nextDouble();

System.out.print("Enter GST (%): ");
```

```
        double tax = sc.nextDouble();



        double afterDiscount = hotel.applyDiscount(grandTotal, discount);

        double finalAmount = hotel.applyTax(afterDiscount, tax);



        // Final Invoice

        System.out.println("\n============ HOTEL BILL ============");

        System.out.println("Customer Name : " + customerName);

        System.out.println("Table Number  : " + tableNo);

        System.out.println("------------------------------");

        System.out.println("Total Amount      : " + grandTotal);

        System.out.println("After Discount    : " + afterDiscount);

        System.out.println("Final Amount (GST): " + finalAmount);

        System.out.println("====================================");

        System.out.println("       THANK YOU! VISIT AGAIN 😊 ");



        sc.close();

    }

}
```

3)MAVEN BUILD

MVN CLEAN TEST

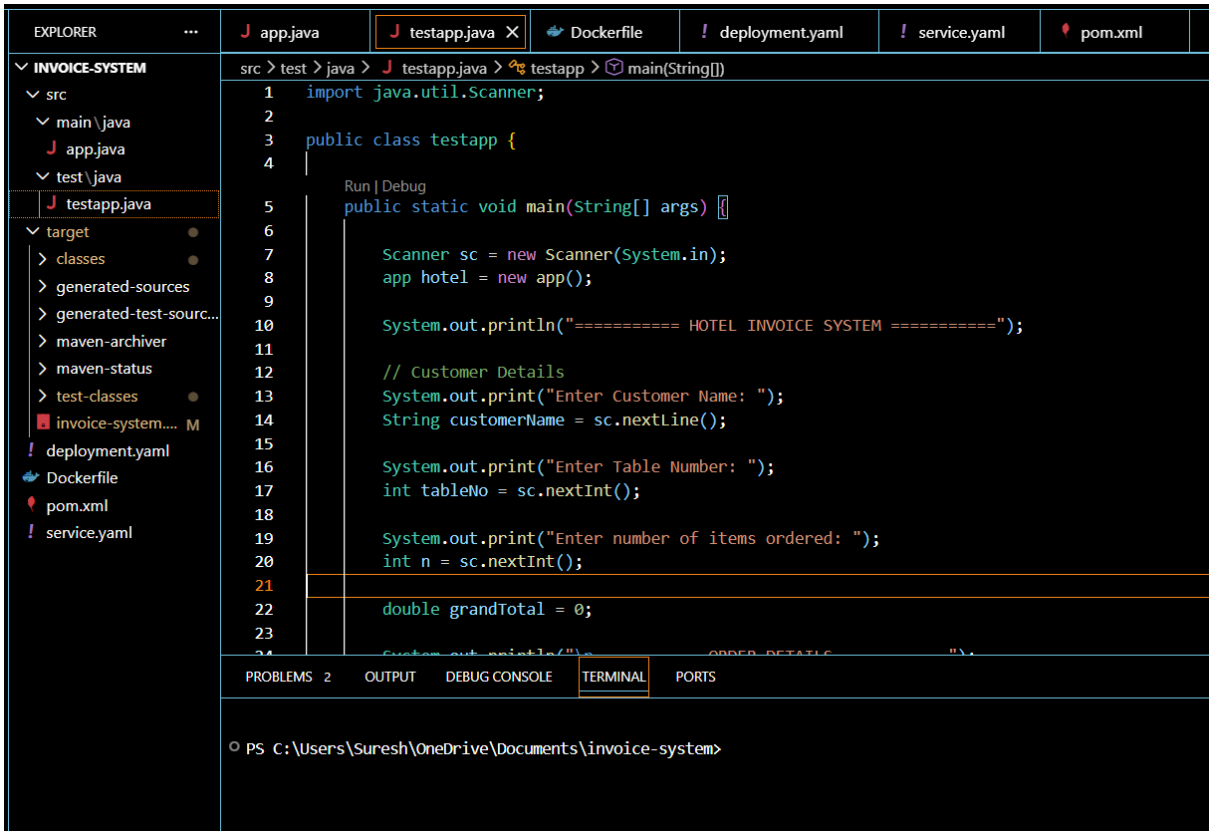
```
PS C:\Users\Suresh\OneDrive\Documents\invoice-system> mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------------< com.example:invoice-system >---------------------
[INFO] Building invoice-system 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ invoice-system ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ invoice-system ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform
 dependent!
[INFO] skip non existing resourceDirectory C:\Users\Suresh\OneDrive\Documents\invoice-system\src\main
\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ invoice-system ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform depen
dent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of syst
em modules automatically
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ invoice-system ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform
 dependent!
[INFO] skip non existing resourceDirectory C:\Users\Suresh\OneDrive\Documents\invoice-system\src\test
\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ invoice-system ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform depen
dent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of syst
em modules automatically
```

MVN CLEAN PACKAGE

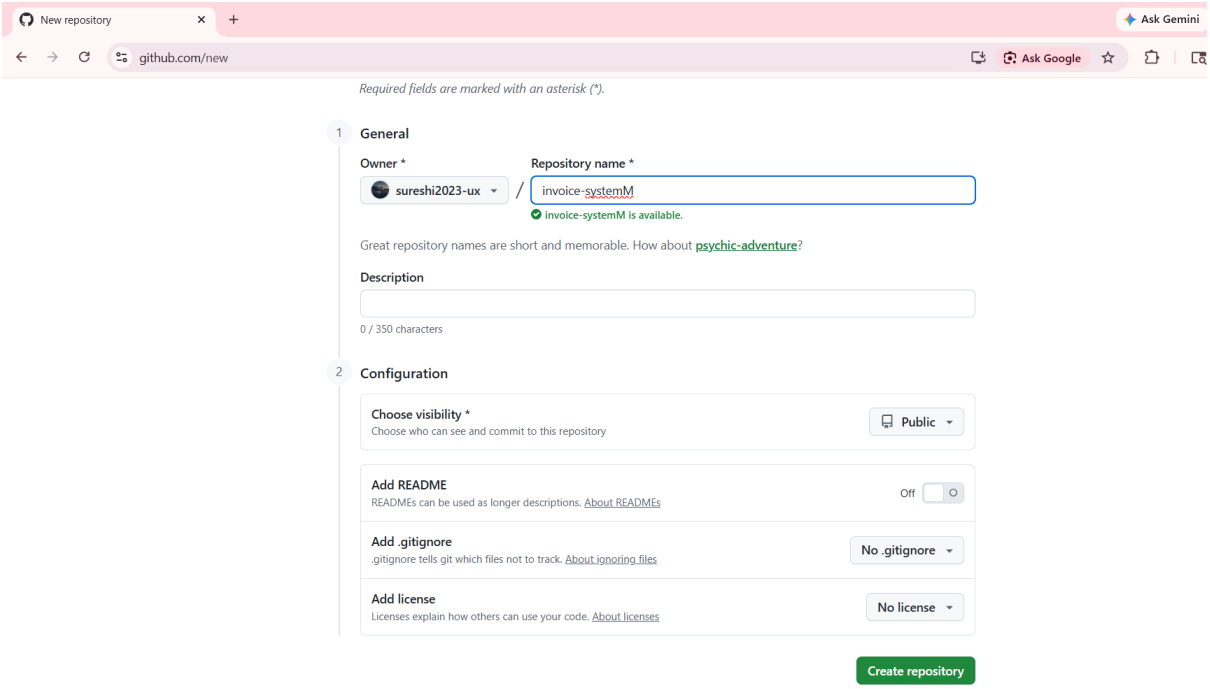
```
PS C:\Users\Suresh\OneDrive\Documents\invoice-system> mvn clean test
[INFO]
[INFO] --- surefire:3.0.0:test (default-test) @ invoice-system ---
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.118 s
[INFO] Finished at: 2026-04-07T12:30:05+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\Suresh\OneDrive\Documents\invoice-system> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:invoice-system >--------------------
[INFO] Building invoice-system 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ invoice-system ---
[INFO] Deleting C:\Users\Suresh\OneDrive\Documents\invoice-system\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ invoice-system ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform
 dependent!
[INFO] skip non existing resourceDirectory C:\Users\Suresh\OneDrive\Documents\invoice-system\src\main
\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ invoice-system ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform depen
dent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of syst
em modules automatically
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ invoice-system ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform
 dependent!
[INFO] skip non existing resourceDirectory C:\Users\Suresh\OneDrive\Documents\invoice-system\src\test
\resources
```

```
 dependent!
[INFO] skip non existing resourceDirectory C:\Users\Suresh\OneDrive\Documents\invoice-system\src\test
\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ invoice-system ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform depen
dent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of syst
em modules automatically
[INFO]
[INFO] --- surefire:3.0.0:test (default-test) @ invoice-system ---
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ invoice-system ---
[INFO] Building jar: C:\Users\Suresh\OneDrive\Documents\invoice-system\target\invoice-system.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.795 s
[INFO] Finished at: 2026-04-07T12:30:17+05:30
[INFO] ------------------------------------------------------------------------
```

TARGET FOLDER CREATED



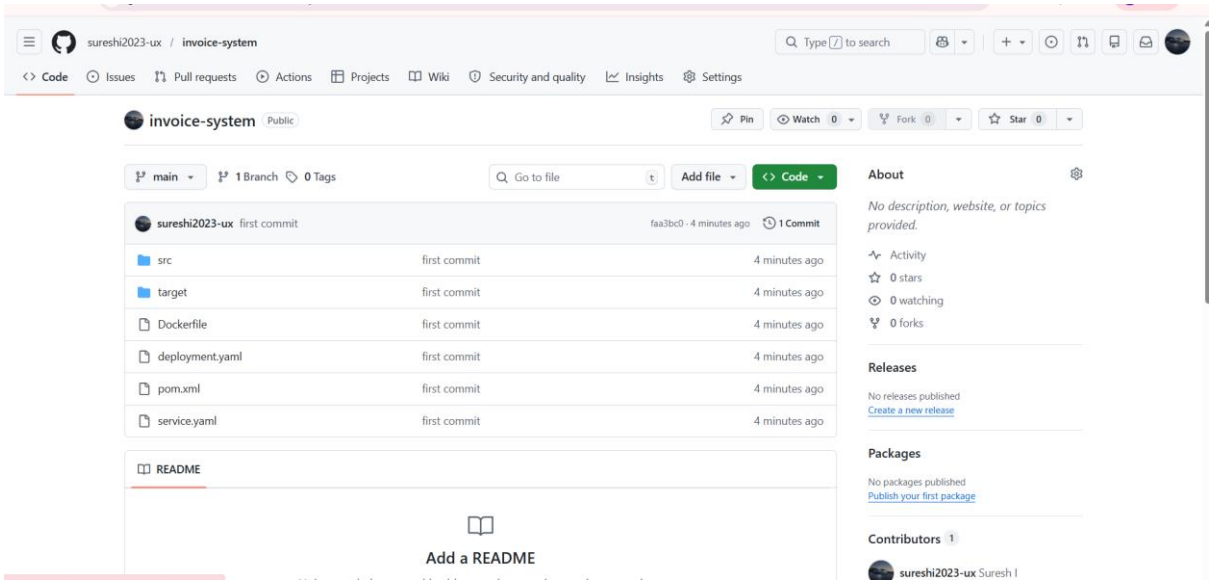
4) CREATE THE REPOSITORY IN GIT


New repository
Ask Gemini
github.com/new
Ask Google
Required fields are marked with an asterisk (*).
1
General
Owner *
Repository name *
sureshi2023-ux
/
invoice-systemM
invoice-systemM is available.
Great repository names are short and memorable. How about psychic-adventure?
Description
0 / 350 characters
2
Configuration
Choose visibility *
Choose who can see and commit to this repository
Public
Add README
READMEs can be used as longer descriptions. About READMEs
Off
Add .gitignore
.gitignore tells git which files not to track. About ignoring files
No .gitignore
Add license
Licenses explain how others can use your code. About licenses
No license
Create repository

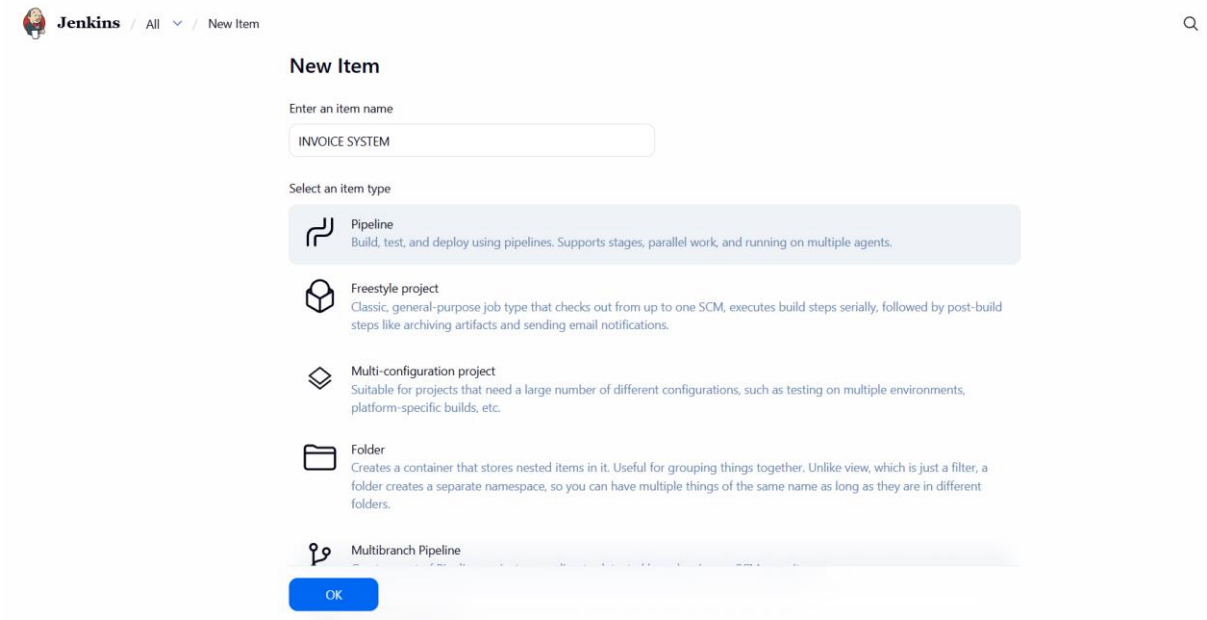
5) PUSH TO GIT

```
PS C:\Users\Suresh\OneDrive\Documents\invoice-system> git init
>> git add README.md
>> git commit -m "first commit"
>> git branch -M main
>> git remote add origin https://github.com/sureshi2023-ux/invoice-system.git
>> git push -u origin main
Reinitialized existing Git repository in C:/Users/Suresh/OneDrive/Documents/invoice-system/.git/
fatal: pathspec 'README.md' did not match any files
On branch main
nothing to commit, working tree clean
error: remote origin already exists.
Enumerating objects: 31, done.
Counting objects: 100% (31/31), done.
Delta compression using up to 16 threads
Compressing objects: 100% (19/19), done.
Writing objects: 100% (31/31), 5.49 KiB | 330.00 KiB/s, done.
Total 31 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/sureshi2023-ux/invoice-system.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Suresh\OneDrive\Documents\invoice-system>
```

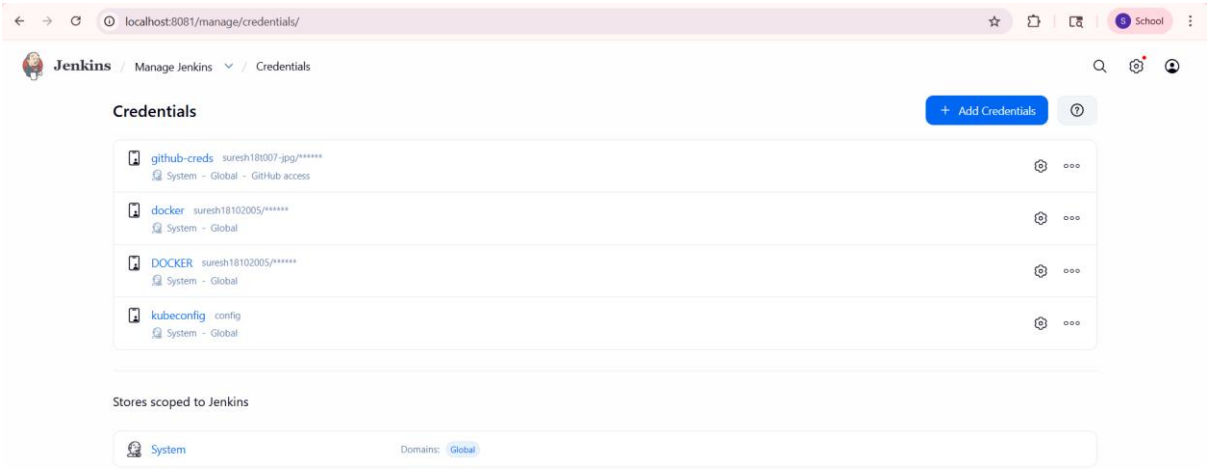
6)FILES ARE UPDATED IN GIT

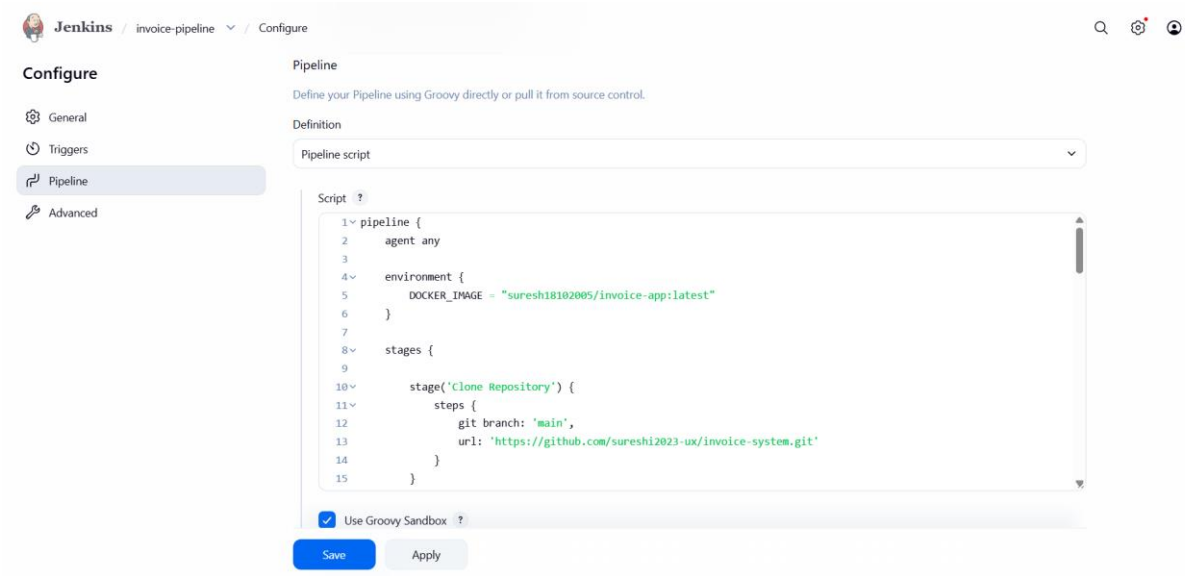


7)CREATE NEW ITEM IN JENKINS

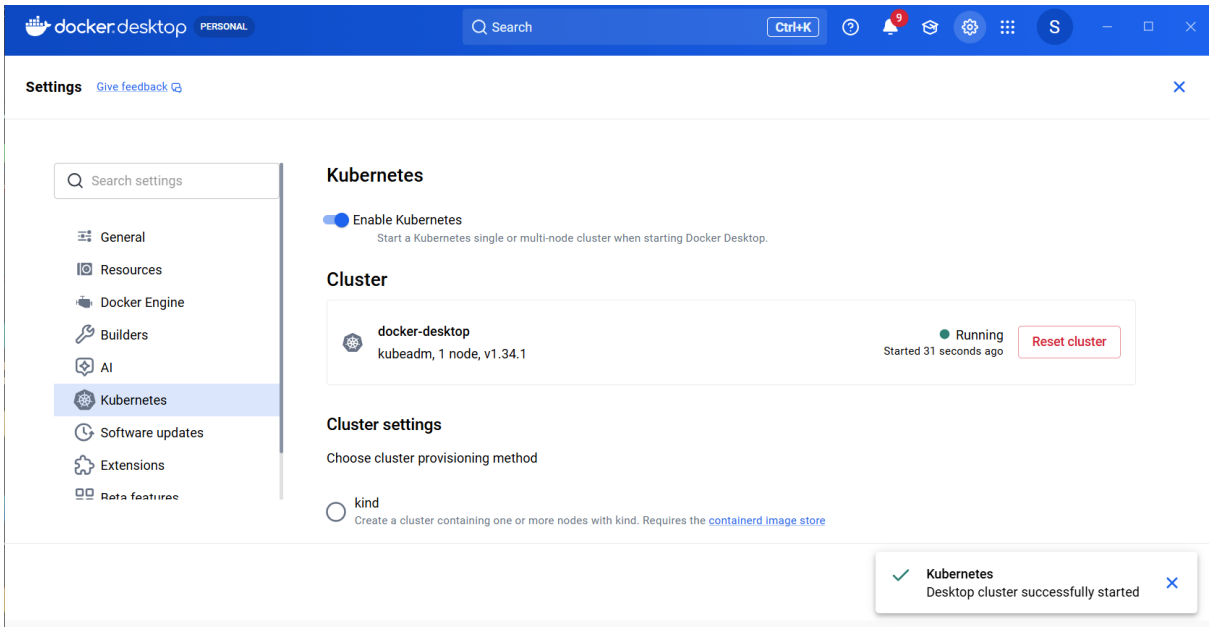


8)CONFIGURE PART ADD PIPELINE CODE AND ADD CREDENTIALS

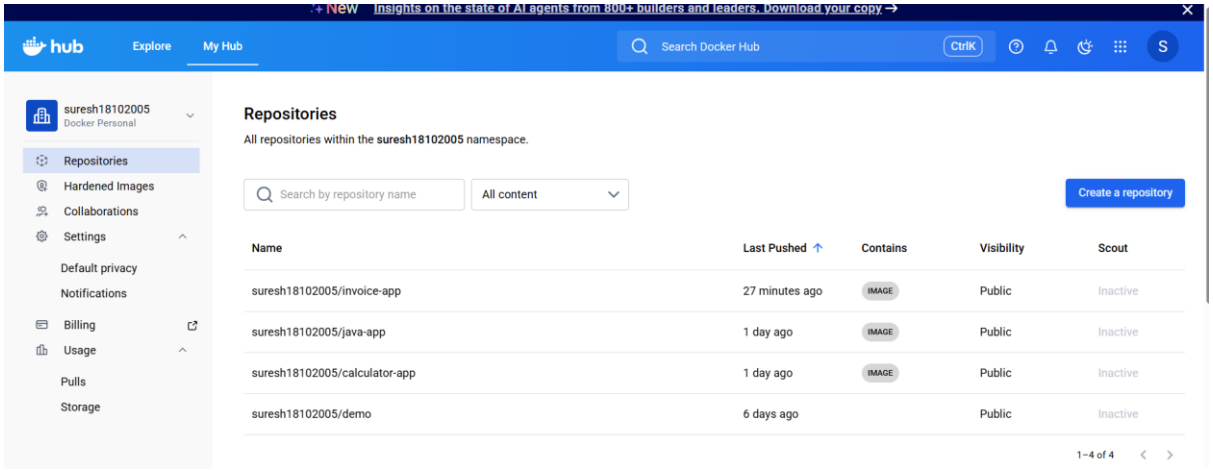




9)MAKE SURE DOCKER IS RUNNING



10) CHECK THE REPOSITORY IS CREATED BY PIPELINE CODE



11)CONSOLE OUTPUT OF JENKINS PIPELINE

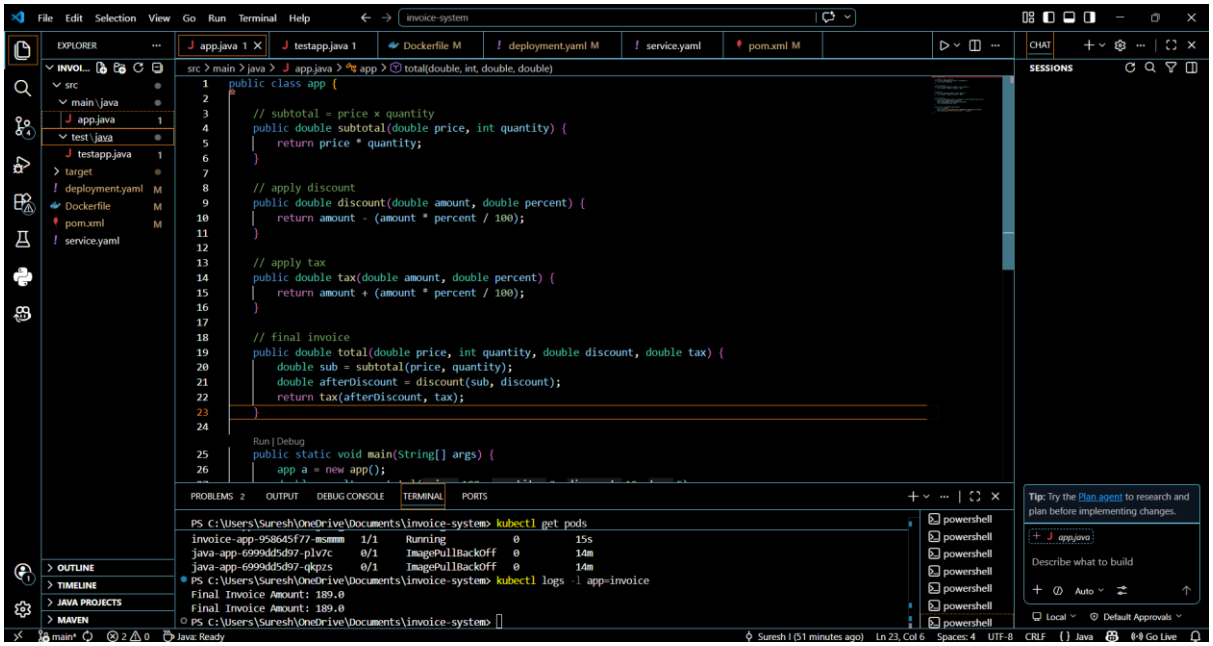
```
C:\ProgramData\Jenkins\.jenkins\workspace\invoice-pipeline>kubectl get pods
NAME                          READY   STATUS             RESTARTS      AGE
invoice-app-958645f77-jb49g   1/1     Running            1 (19h ago)   45h
invoice-app-958645f77-msmmm   1/1     Running            1 (19h ago)   45h
java-app-6999dd5d97-plv7c     0/1     ImagePullBackOff   0             45h
java-app-6999dd5d97-qkpzs     0/1     ImagePullBackOff   0             45h

C:\ProgramData\Jenkins\.jenkins\workspace\invoice-pipeline>kubectl get svc
NAME              TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
invoice-service   NodePort    10.101.108.45   <none>        80:30007/TCP   46h
java-service      NodePort    10.106.182.87   <none>        80:30008/TCP   3d1h
kubernetes        ClusterIP   10.96.0.1       <none>        443/TCP        3d2h
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
✅ Invoice System CI/CD Pipeline executed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

BUILD STAGES COMPLETED

TERMINAL CHECKS

```
PS C:\Users\Suresh\OneDrive\Documents\invoice-system> docker build -t suresh18102005/invoice-app .
[+] Building 2.7s (9/9) FINISHED                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                0.1s
 => => transferring dockerfile: 196B                                                                                0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk-alpine                                   1.9s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io                                             0.0s
 => [internal] load .dockerignore                                                                                   0.1s
 => => transferring context: 2B                                                                                     0.0s
 => [1/3] FROM docker.io/library/eclipse-temurin:17-jdk-alpine@sha256:903e865ecd125a01759fd156d042b998aa86957262f6dec47  0.1s
 => => resolve docker.io/library/eclipse-temurin:17-jdk-alpine@sha256:903e865ecd125a01759fd156d042b998aa86957262f6dec47  0.0s
 => [internal] load build context                                                                                   0.1s
 => => transferring context: 2.64kB                                                                                 0.0s
 => CACHED [2/3] WORKDIR /app                                                                                       0.0s
 => [3/3] COPY target/*.jar app.jar                                                                                 0.0s
 => exporting to image                                                                                              0.3s
 => => exporting layers                                                                                             0.1s
 => => exporting manifest sha256:3238b6a15616aadbdec50dcc57b0a4f43e71fa79d983cf5ad038f170d44b1fdb                   0.0s
 => => exporting config sha256:5742df1016c309c1c6b7034f1d736bb7920e31adc212fbccb5d2c71c6eaad702                     0.0s
 => => exporting attestation manifest sha256:44bb40feb8b2131d3146de402e5b692e70451611e17149bfb01bca9c89f4bc1d       0.0s
 => => exporting manifest list sha256:879fecb13d6d92d7beb2826980983a0fa05fe6c06dca5452d88d008d42004c3c              0.0s
 => => naming to docker.io/suresh18102005/invoice-app:latest                                                        0.0s
 => => unpacking to docker.io/suresh18102005/invoice-app:latest                                                     0.0s
```

```
PS C:\Users\Suresh\OneDrive\Documents\invoice-system> docker push suresh18102005/invoice-app
Using default tag: latest
The push refers to repository [docker.io/suresh18102005/invoice-app]
44f0c932eafa: Pushed
82c540078c28: Layer already exists
7cd2436f004e: Layer already exists
946c51efe0fd: Layer already exists
95ba45b0da8f: Layer already exists
589002ba0eae: Layer already exists
3a547da4fe15: Layer already exists
1ea9421094a6: Pushed
latest: digest: sha256:879fecb13d6d92d7beb2826980983a0fa05fe6c06dca5452d88d008d42004c3c size: 856
```

```
PS C:\Users\Suresh\OneDrive\Documents\invoice-system> kubectl get pods
NAME                          READY   STATUS             RESTARTS   AGE
invoice-app-958645f77-jb49g   1/1     Running            0          15s
invoice-app-958645f77-msmmm   1/1     Running            0          15s
java-app-6999dd5d97-plv7c     0/1     ImagePullBackOff   0          14m
java-app-6999dd5d97-qkpzs     0/1     ImagePullBackOff   0          14m
PS C:\Users\Suresh\OneDrive\Documents\invoice-system> kubectl logs -l app=invoice
Final Invoice Amount: 189.0
Final Invoice Amount: 189.0
```

OUTPUT OF THE APP.JAVA CODE